

Clustering

Outline

- Clustering Introduction
- Partitional Clustering Algorithm
 - K-Means
- Hierarchical Algorithm
 - Agglomerative

Clustering Introduction

Goal: Provide an overview of the clustering problem and introduce some of the basic algorithms

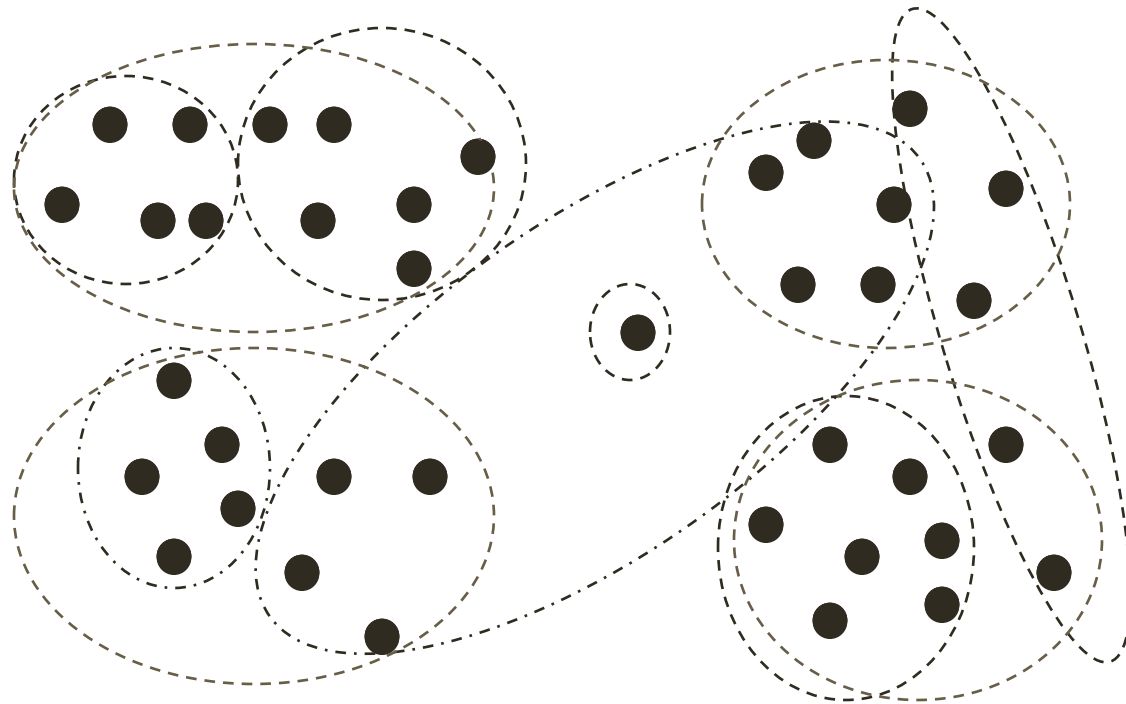
Clustering Examples

- ***Segment*** customer database based on similar buying patterns.
- Group houses in a town into neighborhoods based on similar features.
- Identify new plant species
- Identify similar Web usage patterns

Clustering Example

Income	Age	Children	Marital Status	Education
\$25,000	35	3	Single	High School
\$15,000	25	1	Married	High School
\$20,000	40	0	Single	High School
\$30,000	20	0	Divorced	High School
\$20,000	25	3	Divorced	College
\$70,000	60	0	Married	College
\$90,000	30	0	Married	Graduate School
\$200,000	45	5	Married	Graduate School
\$100,000	50	2	Divorced	College

Clustering Houses



Size Based

Geographic Distance Based

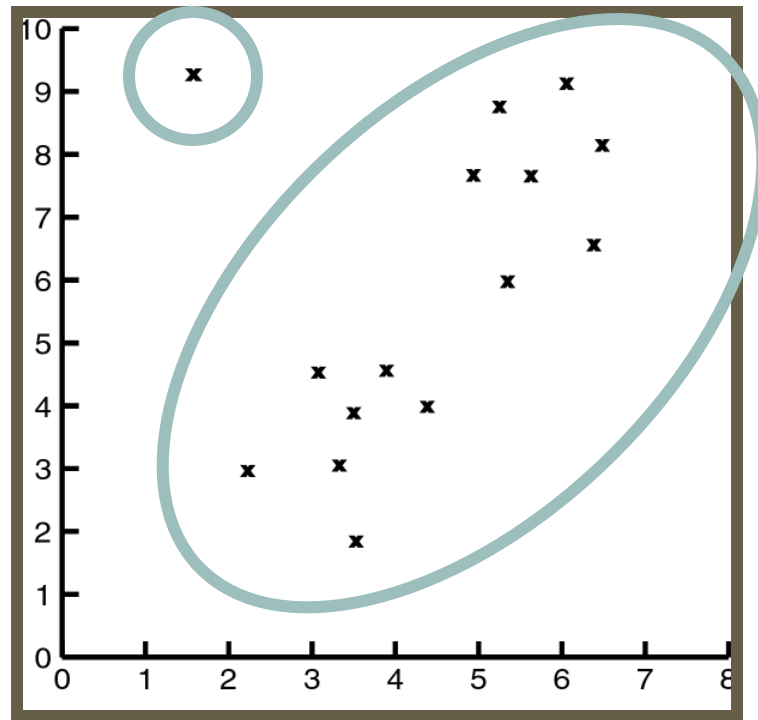
Clustering vs. Classification

- No prior knowledge
 - Number of clusters
 - Meaning of clusters
- Unsupervised learning

Clustering Issues

- Outlier handling
- Dynamic data
- Interpreting results
- Evaluating results
- Number of clusters
- Data to be used
- Scalability

Impact of Outliers on Clustering



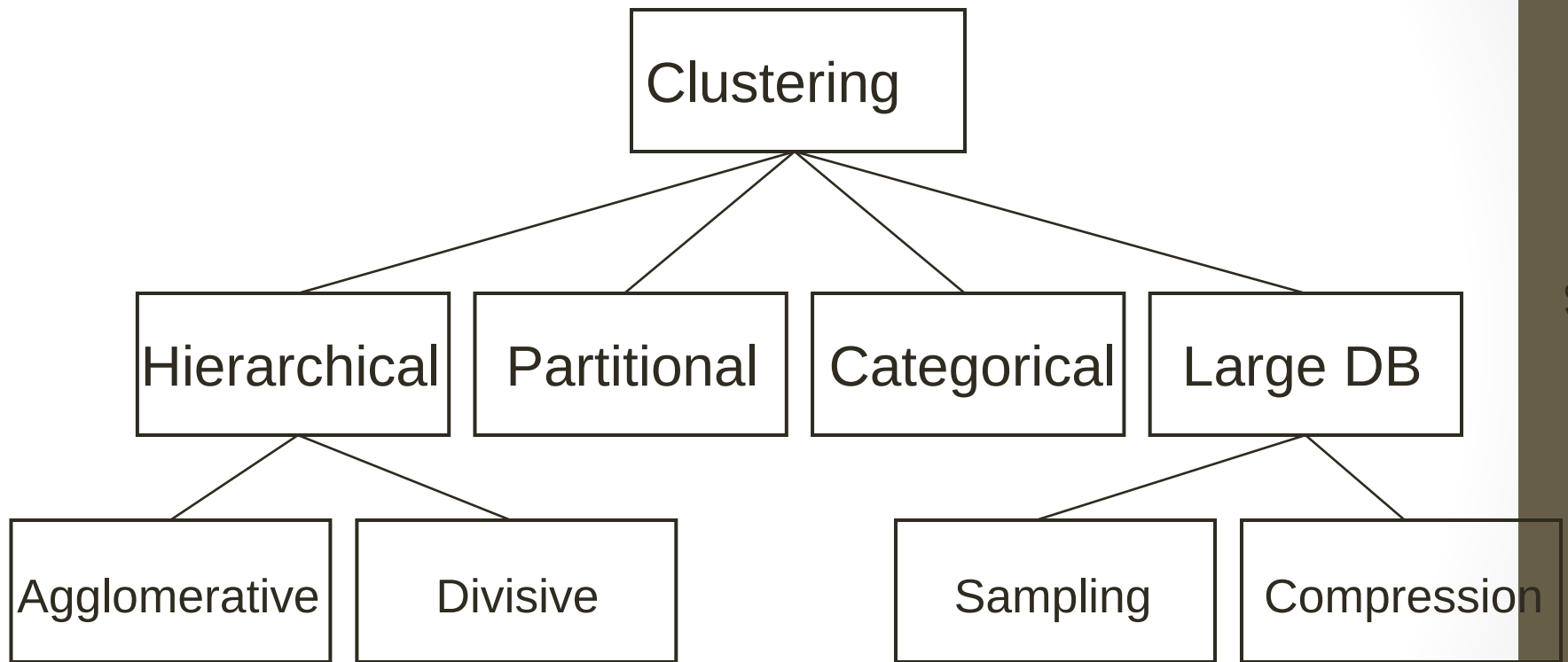
Clustering Problem

- Given a database $D=\{t_1, t_2, \dots, t_n\}$ of tuples and an integer value k , the **Clustering Problem** is to define a mapping $f:D \rightarrow \{1, \dots, k\}$ where each t_i is assigned to one cluster K_j , $1 \leq j \leq k$.
- A **Cluster**, K_j , contains precisely those tuples mapped to it.
- Unlike classification problem, clusters are not known a priori.

Types of Clustering

- *Hierarchical* – Nested set of clusters created.
- *Partitional* – One set of clusters created.
- *Incremental* – Each element handled one at a time.
- *Simultaneous* – All elements handled together.
- *Overlapping/Non-overlapping*

Clustering Approaches



Partitional Clustering

- Nonhierarchical
- Creates clusters in one step as opposed to several steps.
- Since only one set of clusters is output, the user normally has to input the desired number of clusters, k .
- Usually deals with static sets.

Partitional Algorithms

- MST
- Squared Error
- **K-Means**
- Nearest Neighbor
- PAM
- BEA
- GA

Squared Error

- Minimized squared error

$$se_{K_i} = \sum_{j=1}^m \|t_{ij} - C_k\|^2$$

$$se_K = \sum_{j=1}^k se_{K_j}$$

K-Means

- Initial set of clusters randomly chosen.
- Iteratively, items are moved among sets of clusters until the desired set is reached.
- High degree of similarity among elements in a cluster is obtained.
- Given a cluster $K_i = \{t_{i1}, t_{i2}, \dots, t_{im}\}$, the **cluster mean** is $m_i = (1/m)(t_{i1} + \dots + t_{im})$

K-Means Example

- Given: $\{2,4,10,12,3,20,30,11,25\}$, $k=2$
- Randomly assign means: $m_1=3, m_2=4$
- $K_1=\{2,3\}$, $K_2=\{4,10,12,20,30,11,25\}$,
 $m_1=2.5, m_2=16$
- $K_1=\{2,3,4\}$, $K_2=\{10,12,20,30,11,25\}$, $m_1=3, m_2=18$
- $K_1=\{2,3,4,10\}$, $K_2=\{12,20,30,11,25\}$,
 $m_1=4.75, m_2=19.6$
- $K_1=\{2,3,4,10,11,12\}$, $K_2=\{20,30,25\}$, $m_1=7, m_2=25$
- Stop as the clusters with these means are the same.

Hierarchical Agglomerative Clustering (HAC)

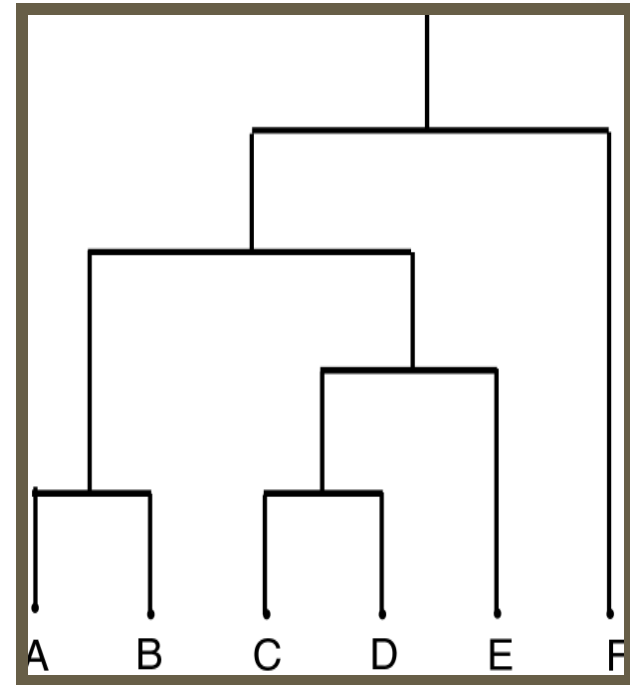
- Starts with one cluster , individual item in its own cluster and iteratively merge clusters until all the items belong to one cluster.
- Bottom up approach is followed to merge the clusters together.
- Dendrograms are pictorially used to represent the HAC.

Different ways to find distance between clusters

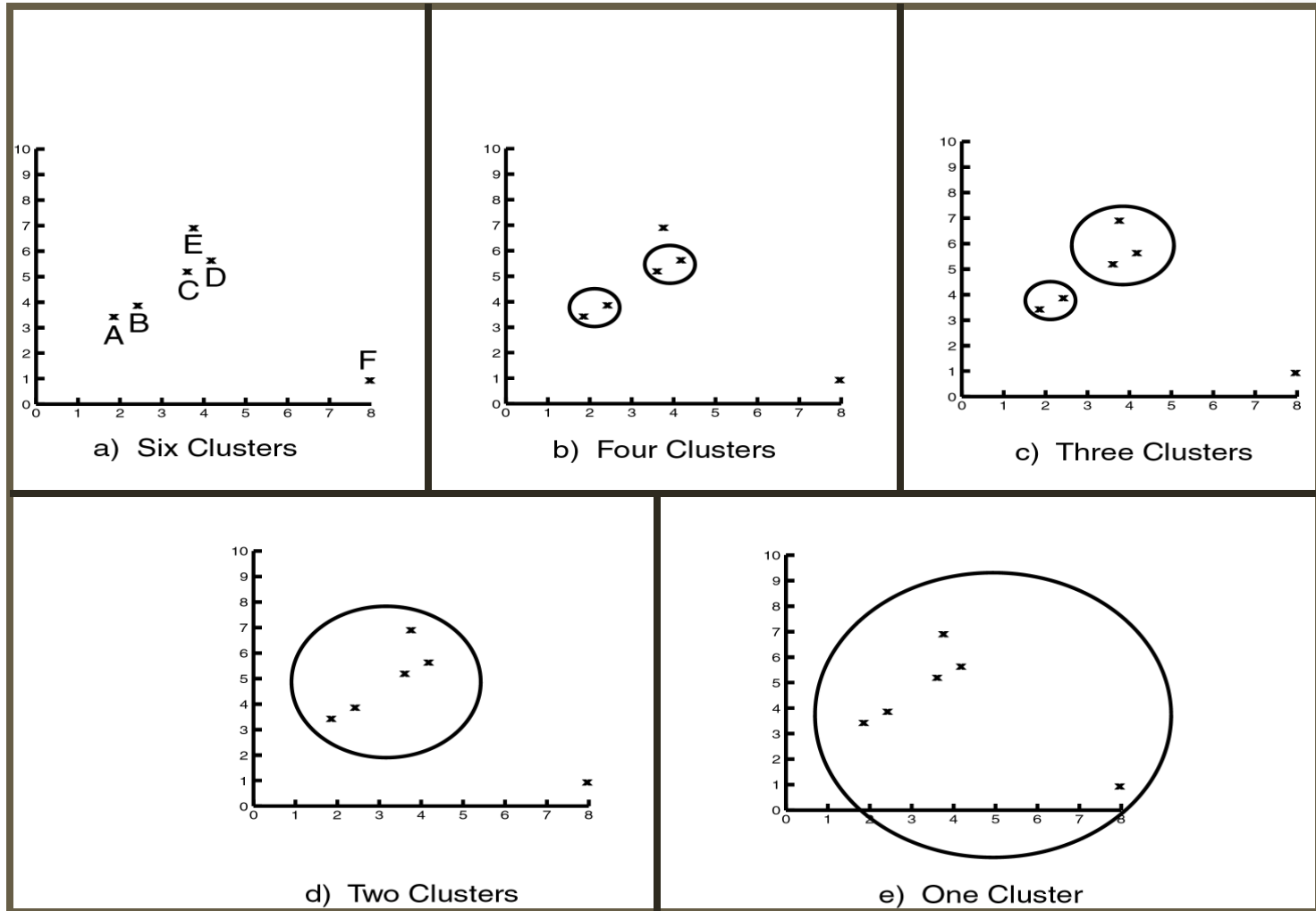
Single Linkage	This is the distance between the closest members of the two clusters.
Complete Linkage	This is the distance between the members that are farthest apart.
Average Linkage	This method involves looking at the distances between all pairs and averages all of these distances. This is also called Unweighted Pair Group Mean Averaging.

Dendrogram

- ***Dendrogram***: a tree data structure which illustrates hierarchical clustering techniques.
- Each level shows clusters for that level.
 - Leaf – individual clusters
 - Root – one cluster
- A cluster at level i is the union of its children clusters at level $i+1$.

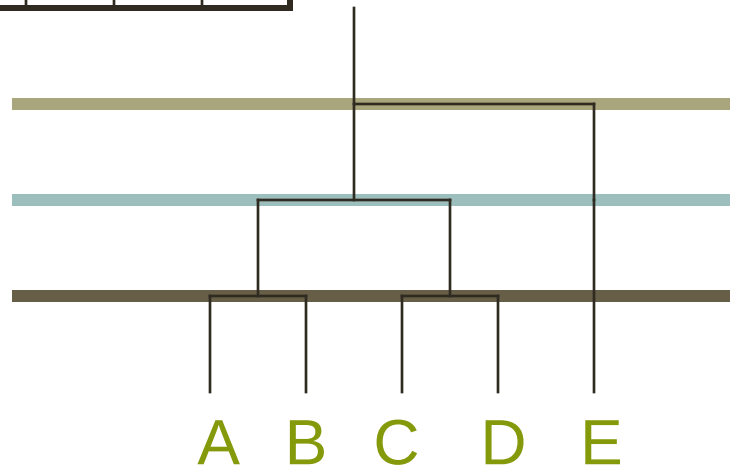
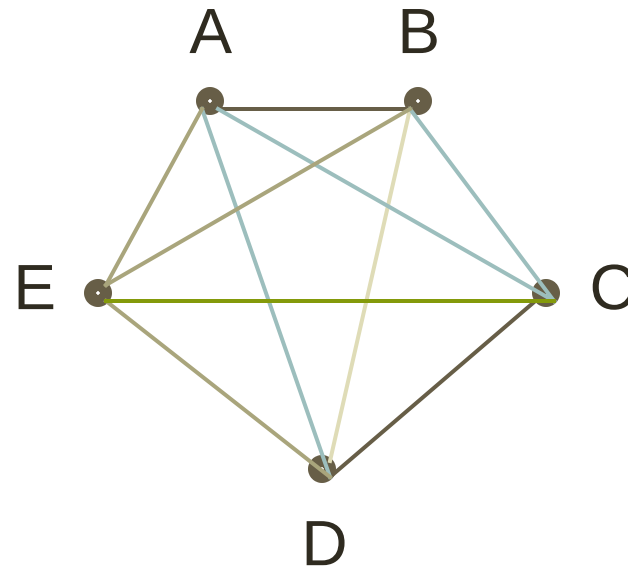


Levels of Clustering



Agglomerative Example

	A	B	C	D	E
A	0	1	2	2	3
B	1	0	2	4	3
C	2	2	0	1	5
D	2	4	1	0	3
E	3	3	5	3	0



Threshold of

1 2 3 4 5

Agglomerative Algorithm

Input:

$D = \{t_1, t_2, \dots, t_n\}$ // Set of elements

A // Adjacency matrix showing distance between elements.

Output:

DE // Dendrogram represented as a set of ordered triples.

Agglomerative Algorithm:

$d = 0;$

$k = n;$

$K = \{\{t_1\}, \dots, \{t_n\}\};$

$DE = \{< d, k, K >\};$ // Initially dendrogram contains each element in its own cluster

repeat

$oldk = k;$

$d = d + 1;$

$A_d =$ Vertex adjacency matrix for graph with threshold distance of d ;

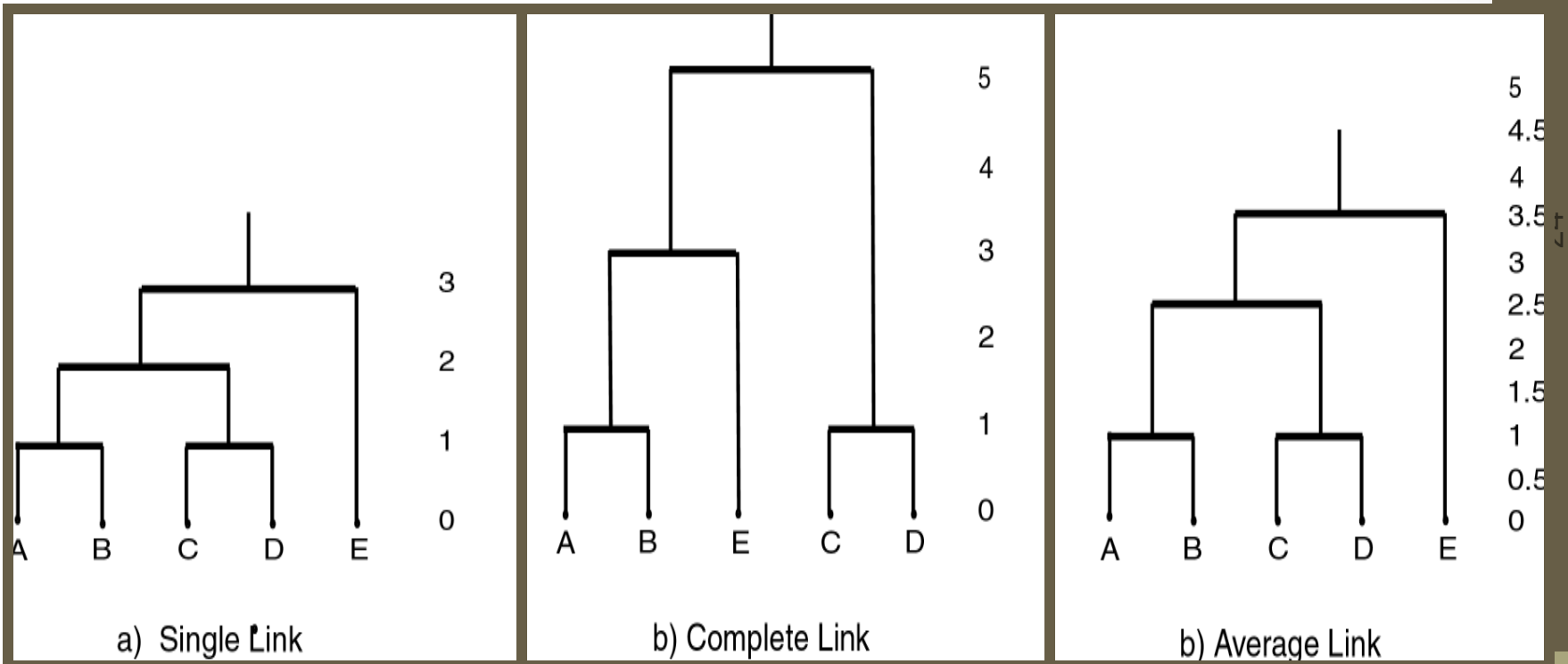
$< k, K > = NewClusters(A_d, D);$

if $oldk \neq k$ **then**

$DE = DE \cup < d, k, K >;$ // New set of clusters added to dendrogram.

until $k = 1$

Single Link, Complete, Average, Clustering



Example

Find the clusters using complete link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

- Calculate Euclidean distance, create the distance matrix.

$$\text{Distance } [(x,y), (a,b)] = \sqrt{(x - a)^2 + (y - b)^2}$$

$$\text{Distance (P1,P2)} = \sqrt{(0.40 - 0.22)^2 + (0.53 - 0.38)^2}$$

$$(0.40,0.53), (0.22,0.38) = \sqrt{(0.18)^2 + (0.15)^2}$$

$$= \sqrt{0.0324 + 0.0225}$$

$$= \sqrt{0.0549}$$

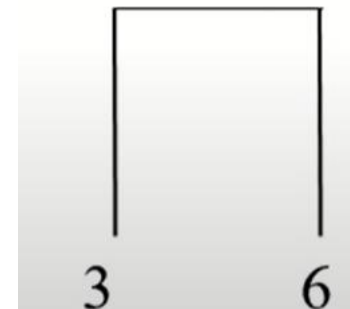
$$= 0.23$$

The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

Distance matrix

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0



The updated distance matrix for cluster P3,P6

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.23	0.25	0		
P4	0.37	0.20	0.22	0	
P5	0.34	0.14	0.39	0.29	0

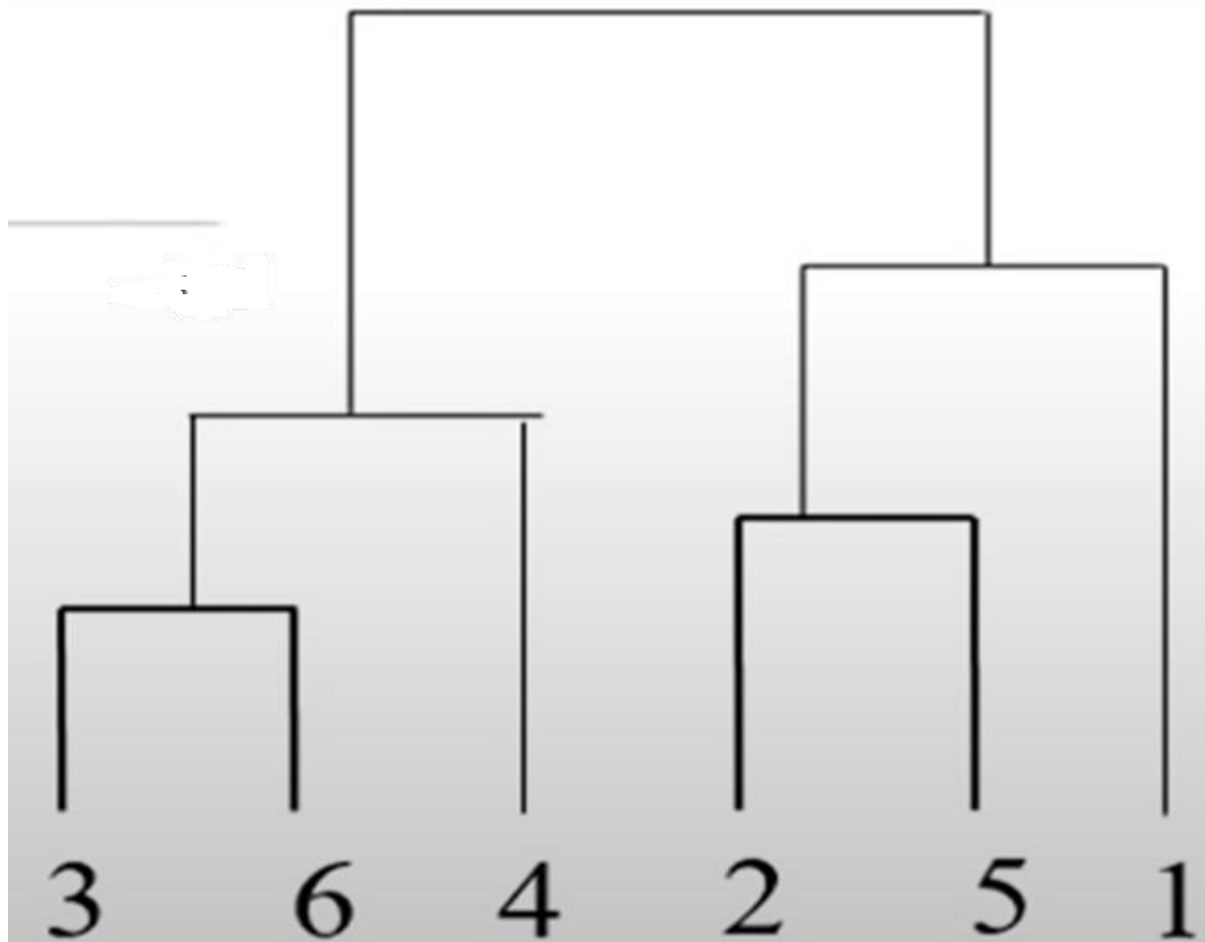
To update the distance matrix $\text{MAX}[\text{dist}(\text{P3,P6}), \text{P1}]$

$\text{MAX}(\text{dist}(\text{P3,P1}), (\text{P6,P1}))$

$= \text{MAX}[(0.22, 0.23)]$

$= 0.23$

Final Dendrogram



Quiz with answers

1. Which among the following is/are some of the assumptions made by the k-means algorithm (assuming Euclidean distance measure)?
 - (a) Clusters are spherical in shape
 - (b) Clusters are of similar sizes
 - (c) Data points in one cluster are well separated from data points of other clusters
 - (d) There is no wide variation in density among the data points

Sol. (a) & (b)

The Euclidean distance measure ensures that areas around a cluster centroid comprising points closest to that centroid (which is a cluster) is spherical in shape. Also, this particular distance measure prevents arbitrarily sized clusters since this typically violates the clustering criterion.

Quiz with answers

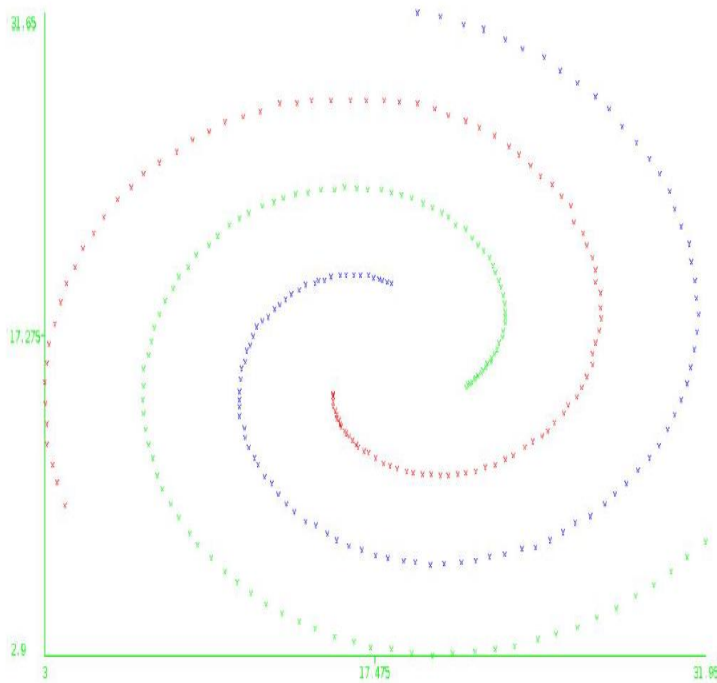
4. Considering single-link and complete-link hierarchical clustering, is it possible for a point to be closer to points in other clusters than to points in its own cluster? If so, in which approach will this tend to be observed?
- (a) No
 - (b) Yes, single-link clustering
 - (c) Yes, complete-link clustering
 - (d) Yes, both single-link and complete-link clustering

Sol. (d)

This is possible in both single-link and complete-link clustering. In the single-link case, an example would be two parallel chains where many points are closer to points in the other chain/cluster than to points in their own cluster. In the complete-link case, this notion is more intuitive due to the clustering constraint (measuring distance between two clusters by the distance between their farthest points).

. Consider the following image showing data points belonging to three different clusters (indicated by the colours of the points). Which among the following clustering algorithms will perform well in accurately clustering the given data?

- (a) K-means
- (b) Single-link hierarchical
- (c) Complete-link hierarchical



Ans:

K-means clustering will not work here. Complete link clustering cannot recover the desired clustering since the clusters are spherical in shape.

Single link clustering is ideally suited

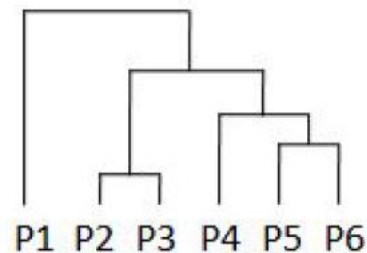
Exercise

Consider the similarity matrix given below.

	P1	P2	P3	P4	P5	P6
P1	1.00	0.70	0.65	0.40	0.20	0.05
P2	0.70	1.00	0.95	0.70	0.50	0.35
P3	0.65	0.95	1.00	0.75	0.55	0.40
P4	0.40	0.70	0.75	1.00	0.80	0.65
P5	0.20	0.50	0.55	0.80	1.00	0.85
P6	0.05	0.35	0.40	0.65	0.85	1.00

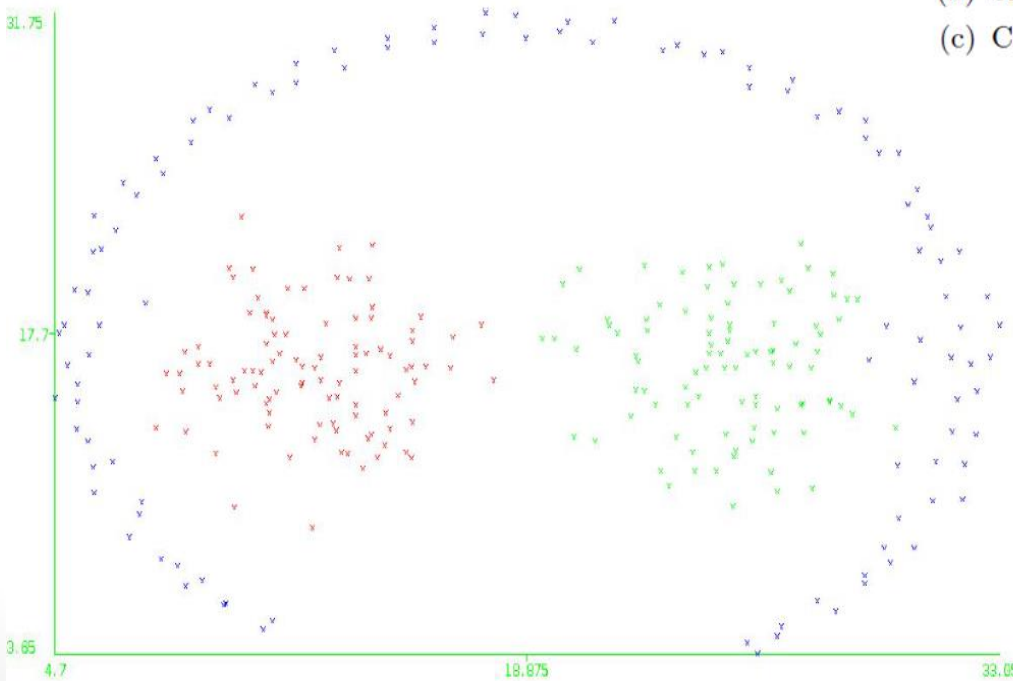
Show the hierarchy of clustering created by the single-link clustering algorithm.

Ans:



- Consider the following image showing data points belonging to three different clusters (indicated by the colours of the points). Which among the following clustering algorithms will perform well in accurately clustering the given data?

- (a) K-means
- (b) Single-link hierarchical
- (c) Complete-link hierarchical



Ans: **None of the mentioned**
K-means and complete-link hierarchical clustering do not do well as explained previously.

The problem with single-link clustering is that there are a few points which belong to different clusters but are close enough to each other so that it may result in combining different clusters.